



Designed and Created By:
Aleisha Vrooman
Class of: 2005

Statement of Scope

Customer Support System

Problem Description

Kids JukeBox USA knows that in order to remain competitive with personalized CD's for children, they must have a working webpage that allows users to order available merchandise. Specifically, there must be a user friendly website in order for the users to search easily and to order their product with a reliable database. Also, the variety of product and customized systems should be considered for the customer. We need to create a system which lets the user choose from a variety of different CD's with a variety of names to insert into each CD. This system also needs to allow the customer to input their personal information so that shipping and billing is possible. Therefore a working, user friendly, web page needs to be edited, and a working database needs to be created and coded into the web page.

Anticipated Business Benefits

The primary long-term business benefit to be obtained from the new system is for Kids JukeBox USA to gain business through accurate and timely cost analysis and adjustment. Additionally, this system will help the company better assist their customers with an easy to understand order form that inputs their information along with what products they wish to purchase. Some of the shorter term goals and benefits which will combine to accomplish the long-term goals are:

- Provide accurate inventory data
- Allow customers to browse through the product information and order the product with ease.

System Capabilities

- Simultaneous multiple user access across a LAN
- 100% data reliability
- Ability for users to purchase products using a web based database.
- View a Bill of Materials for any given product
- Provide an interface where users can experiment with financial parameters (without changing the quantities/costs/prices in inventory)
- Print reports to include: Bill of Materials, Reverse Bill of Material and Margin Analysis

Project Charter

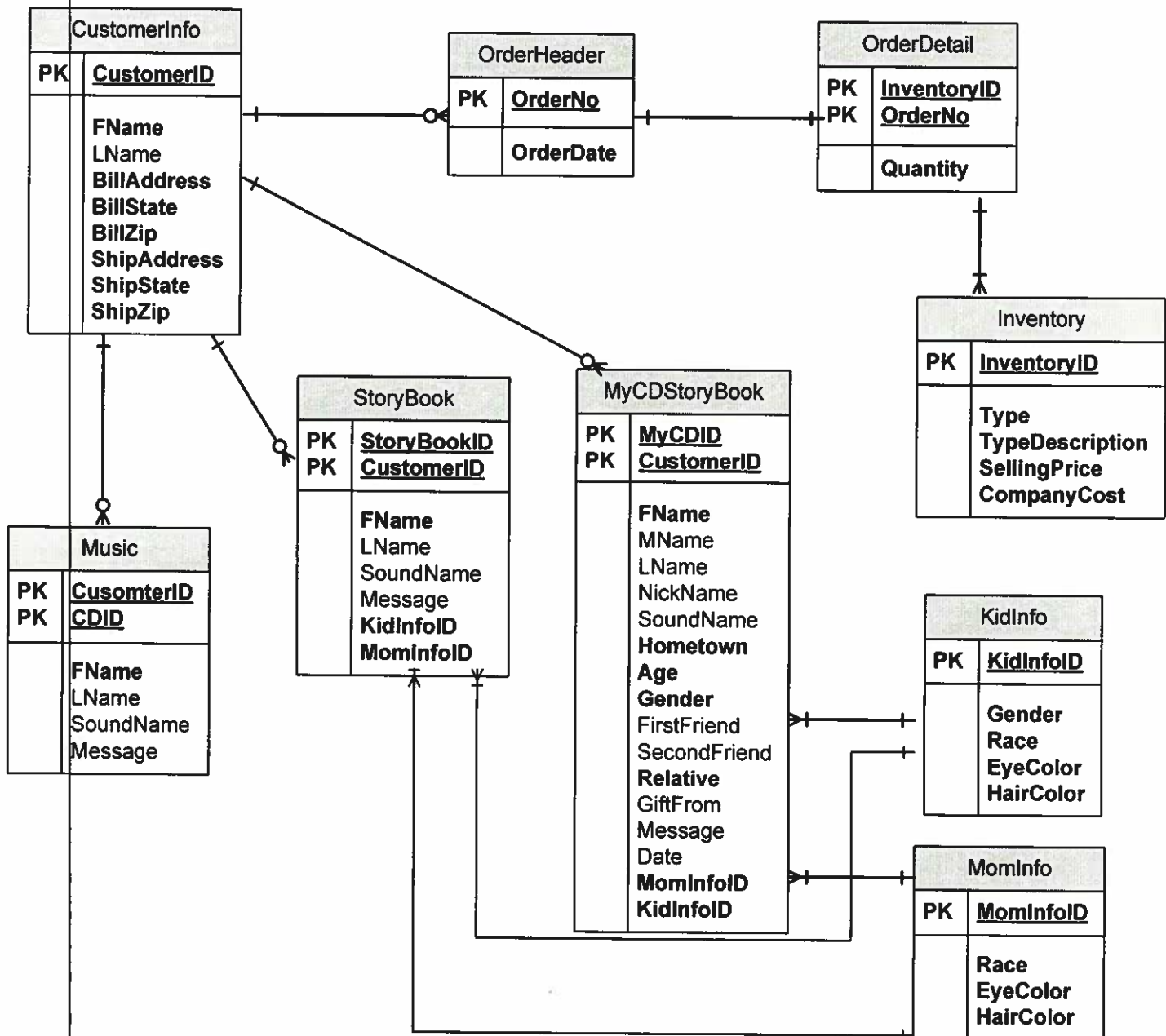
Project Name: Kids JukeBox USA Web Page and Database

Project Purpose: Kids JukeBox USA knows that in order to remain competitive with personalized CD's for children, they must have a working webpage that allows users to order available merchandise. Specifically, there must be a user friendly website in order for the users to search easily and to order their product with a reliable database. Also, the variety of product and customized systems should be considered for the customer. I will create a system which lets the user choose from a variety of different CD's, storybooks, etc., with a variety of names to insert into each CD. This system will also allow input of the customer's personal information so that shipping and billing is possible.

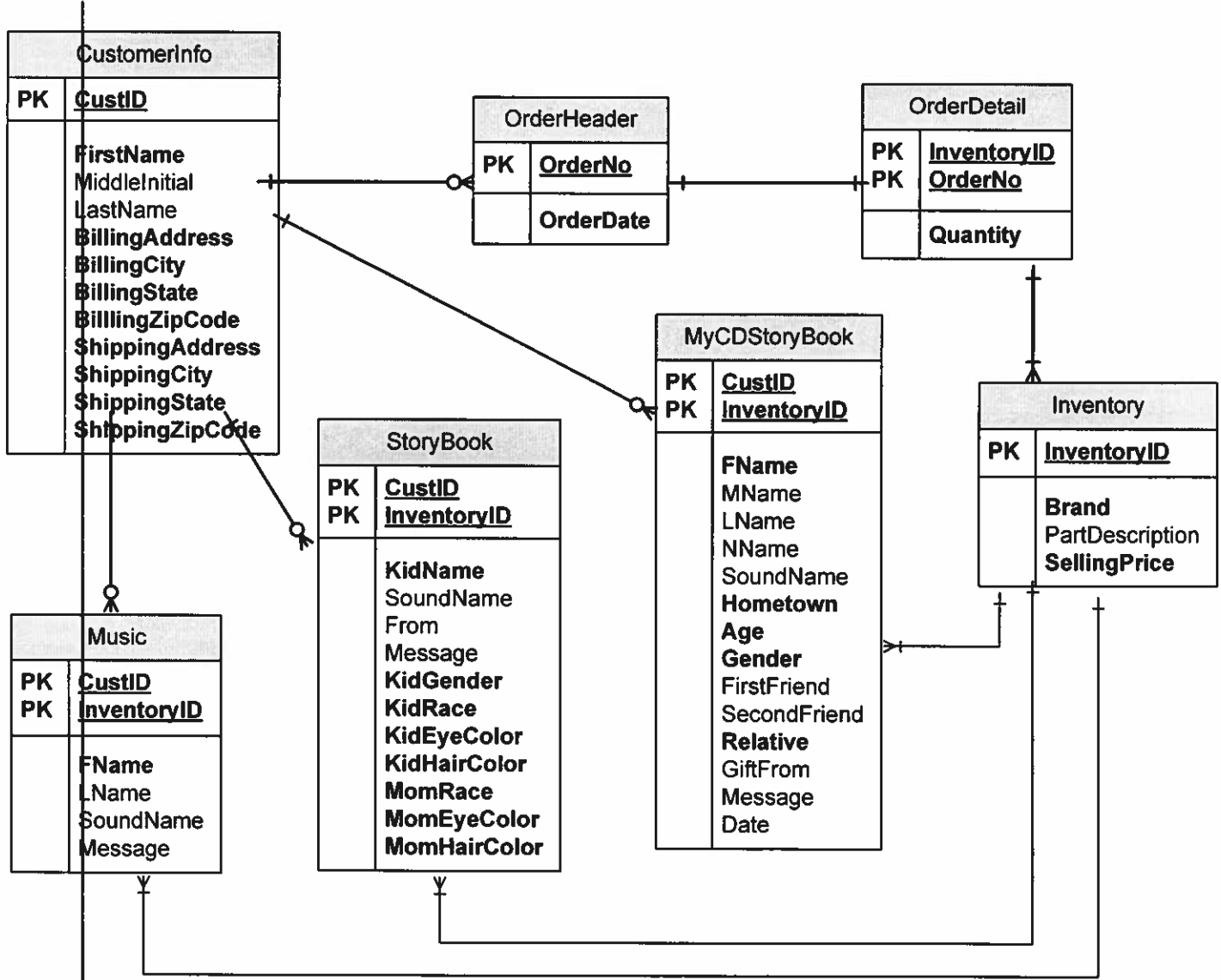
Project Goals: To provide real-time accurate inventory data and to allow customers to build\customize a computer cheaper and faster than our competitors.

Anticipated Completion: Within 8 months of project initiation

Participant	Position	Primary Responsibilities
David Scarborough	Owner of Kids JukeBox USA	To critique and advise the projects progress
Aleisha Vrooman	Programmer and Designer	To design and program the working database based on the needs of David Scarborough



Entity-Relationship Diagram



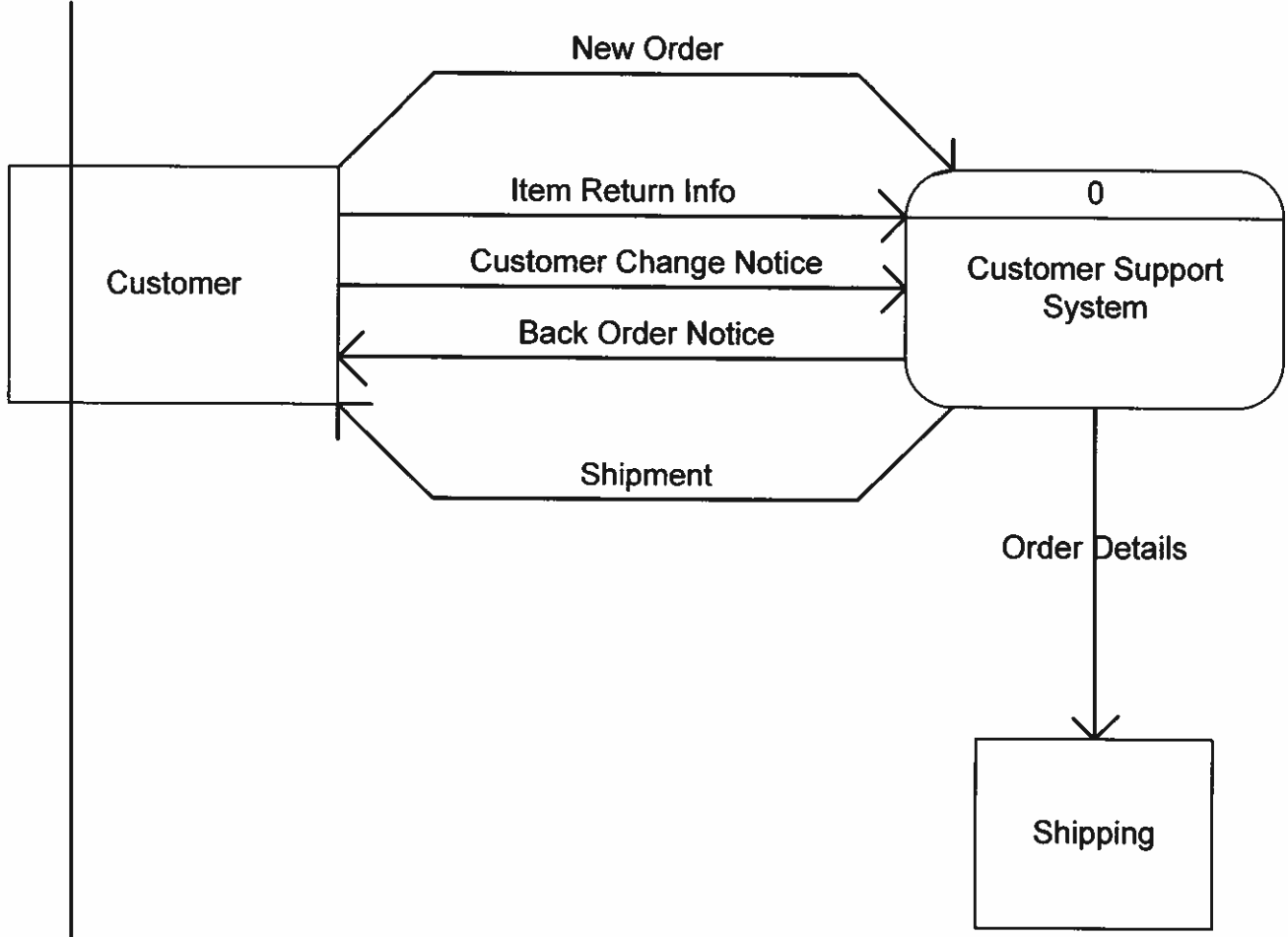
Entity-Relationship Diagram
Revision Feb 2005

ERD With Primary Keys and Foreign Keys

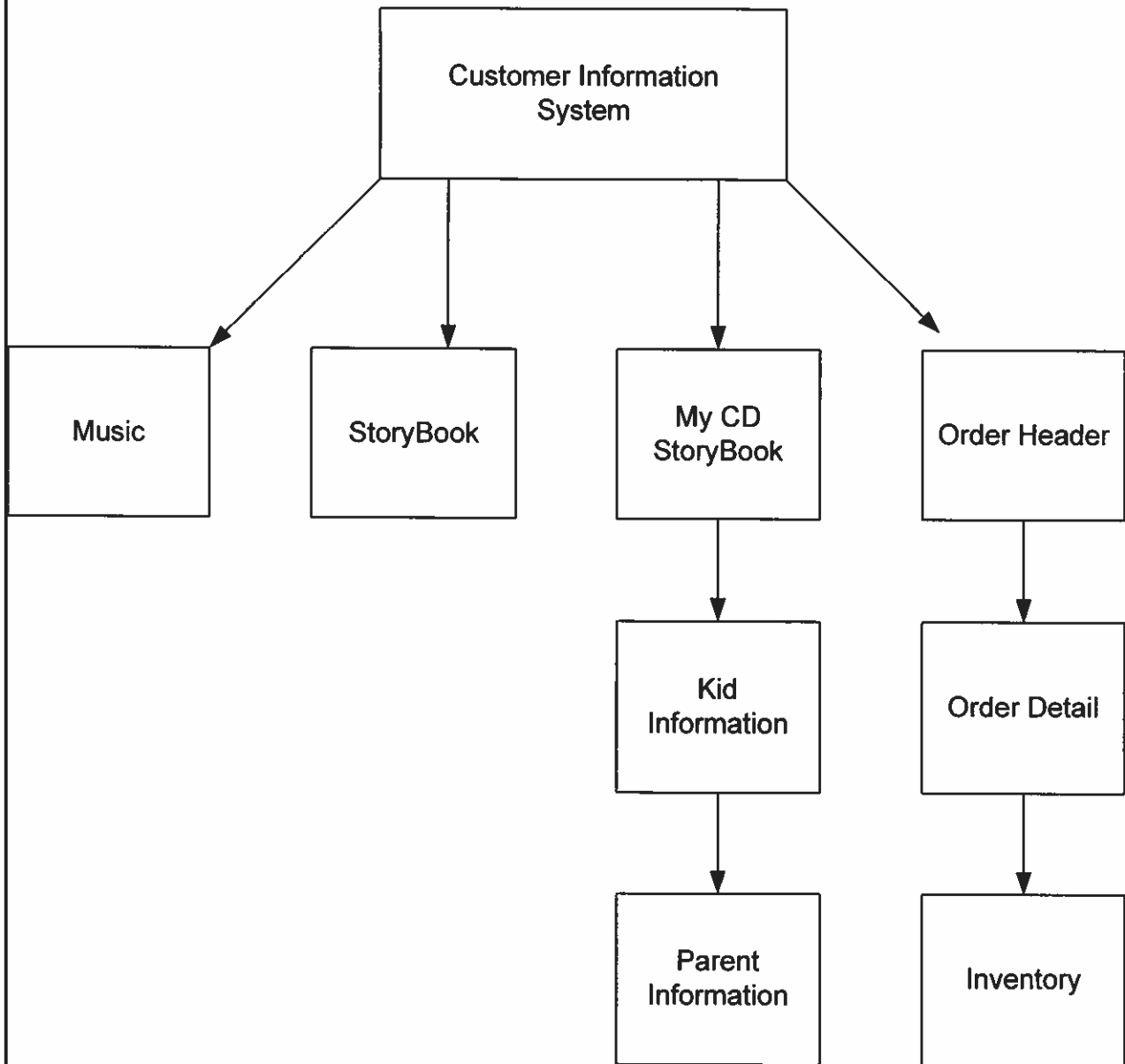
Table	Attributes
CustomerInfo	CustomerID, FName, LName, BillAddress, BillState, BillZip, ShipAddress, ShipState, ShipZip
OrderHeader	OrderNo, CustomerID, OrderDate
OrderDetail	InventoryID, OrderNo, Quantity
Inventory	InventoryID, Type, TypeDescription, SellingPrice, CompanyCost
Music	CDID, CustomerID, FName, LName, SoundName, Message
StoryBook	StoryBookID, CustomerID, FName, LName, SoundName, Message, KidInfoID, MomInfoID
MyCDStoryBook	MyCDID, CustomerID, FName, MName, LName, NName, SName, Town, Age, 1Friend, 2Friend, Relative, Gift, Mge, Date, MomInfoID, KidInfoID
KidInfo	KidInfoID, Gender, Race, EyeColor, HairColor
MomInfo	MomInfoID, Race, EyeColor, HairColor

ERD With Primary Keys and Foreign Keys

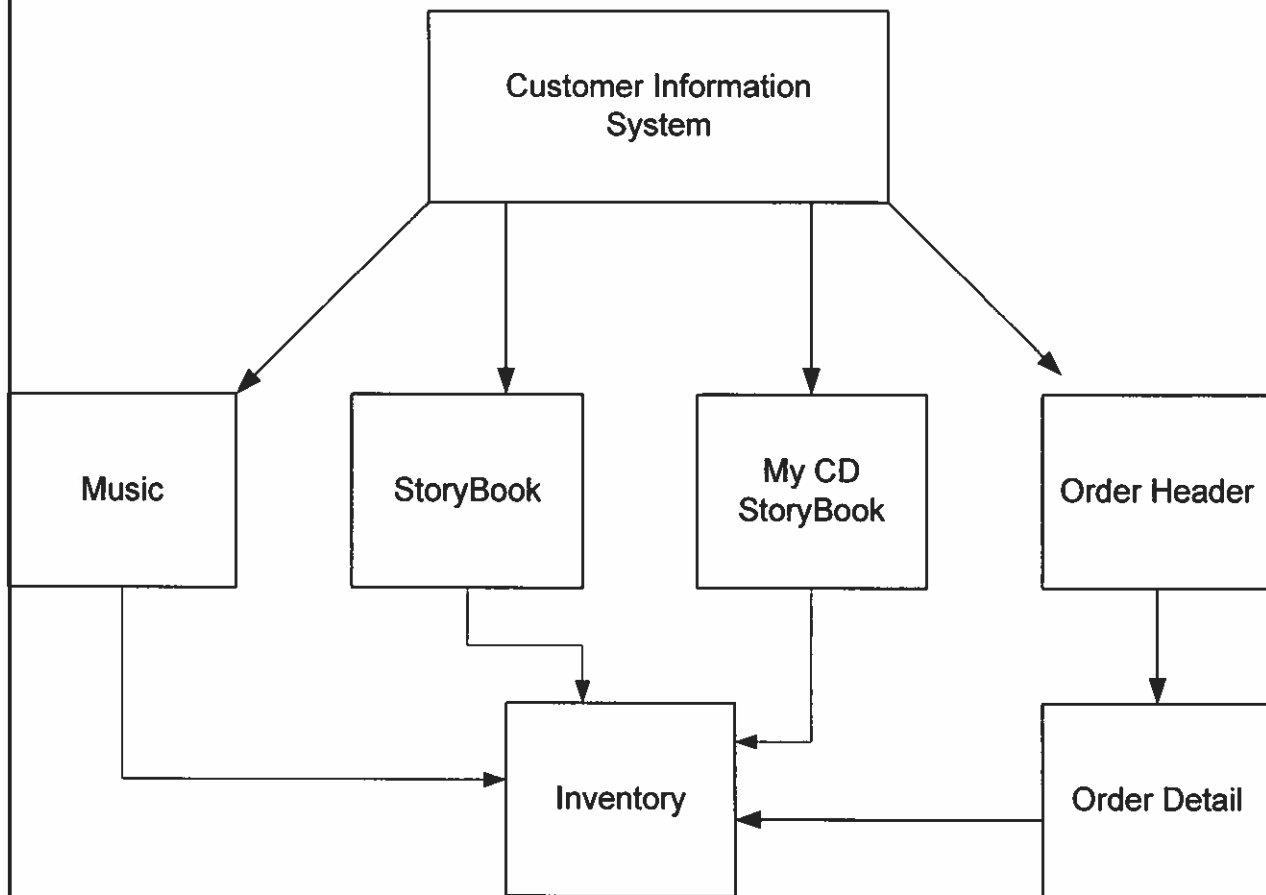
Table	Attributes
Customers	CustID, FirstName, MiddleInitial, LastName, BillingAddress, BillingCity, BillingState, BillingZipCode, ShippingAddress, ShippingCity, ShippingState, ShippingZipCode
OrderHeader	OrderNo, CustID, OrderDate
OrderDetail	InventoryID, OrderNo, Quantity
Inventory	InventoryID, Brand, PartDescription, SellingPrice
Music	InventoryID, CustID, FName, LName, SoundName, Message
StoryBook	InventoryID, CustID, KidName, SoundName, From, Message, KidGender, KidRace, KidEyeColor, KidHairColor, MomRace, MomEyeColor, MomHairColor
MyCDStoryBook	InventoryID, CustID, FName, MName, LName, NName, SoundName, Hometown, Age, Gender, FirstFriend, SecondFriend, Relative, GiftFrom, Message, Date



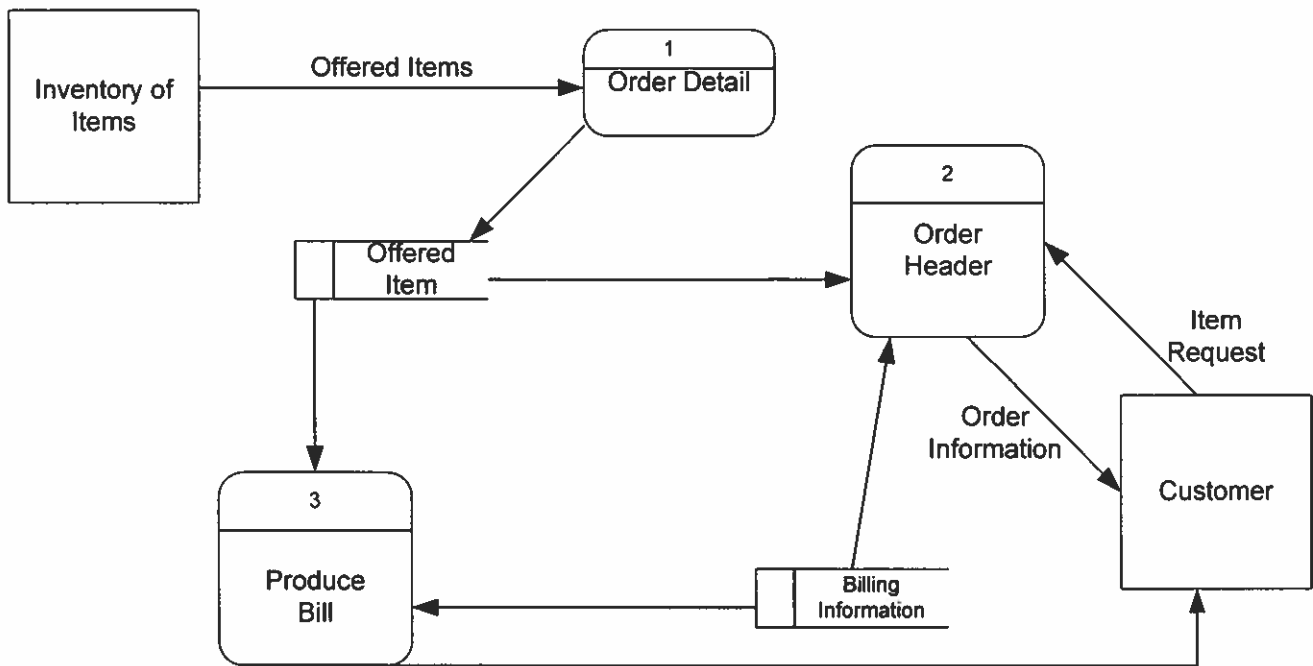
System Context Diagram



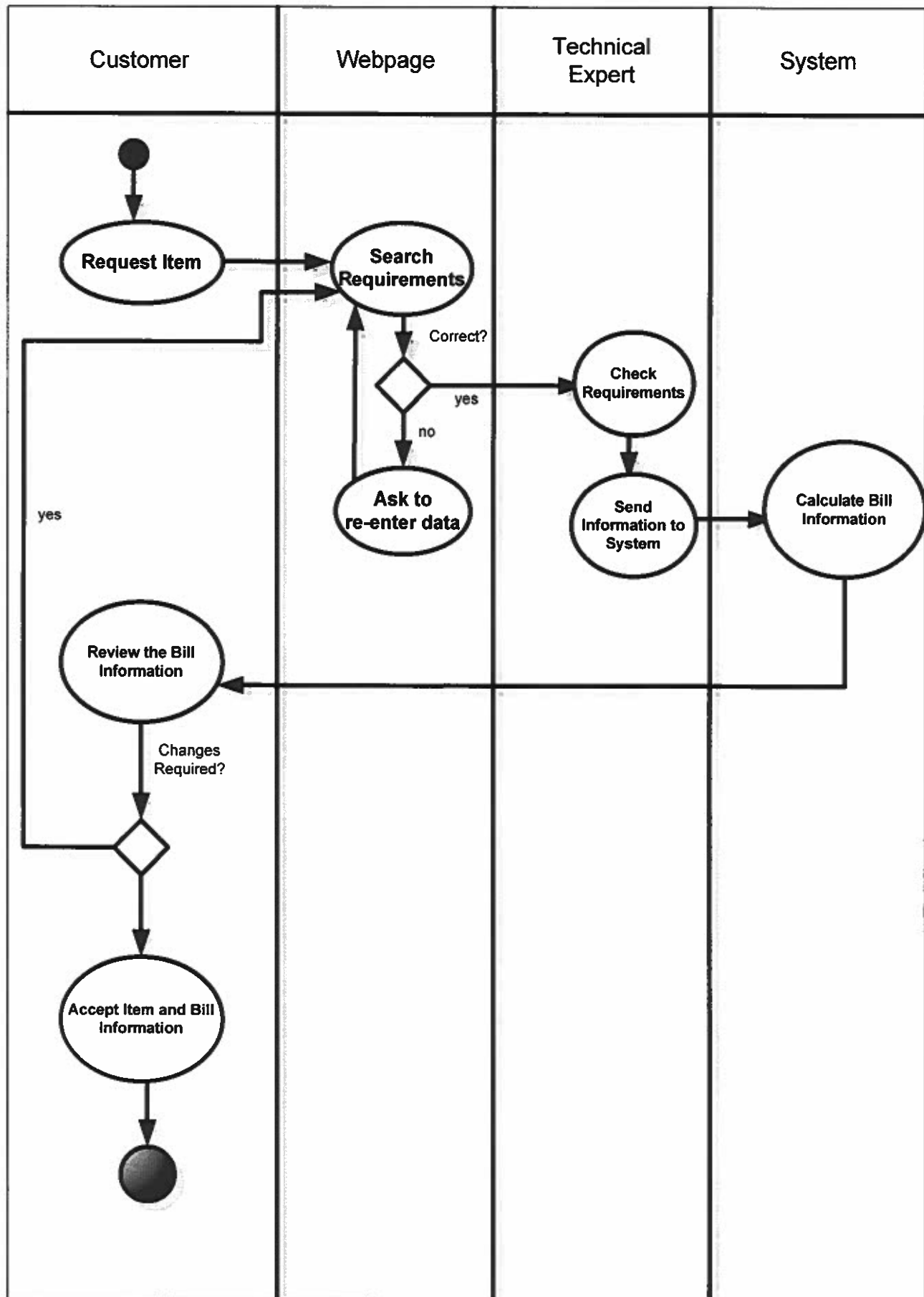
Structure Chart



Structure Chart
Revision Feb 2005



Data Flow Diagram

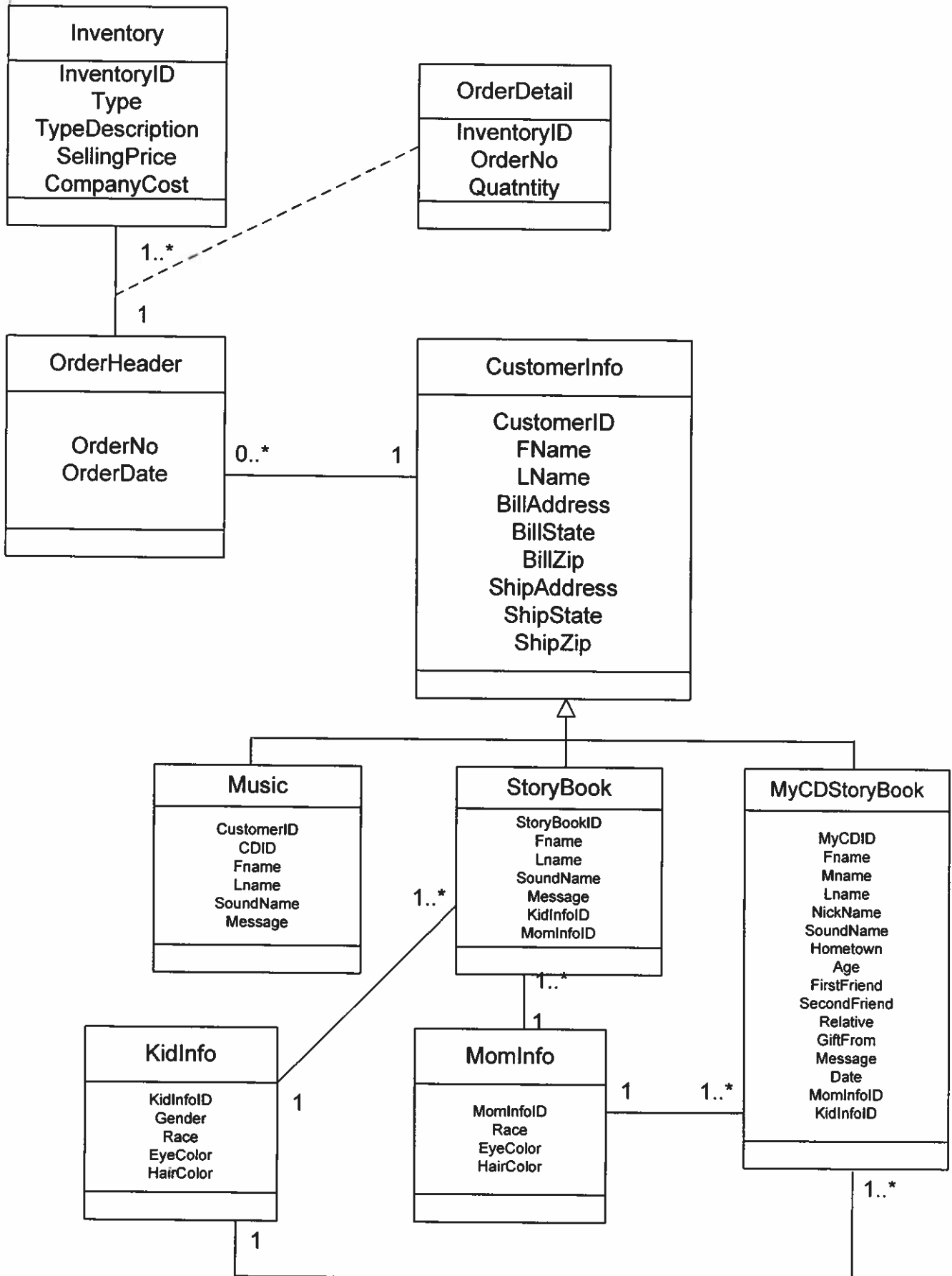


Activity Diagram

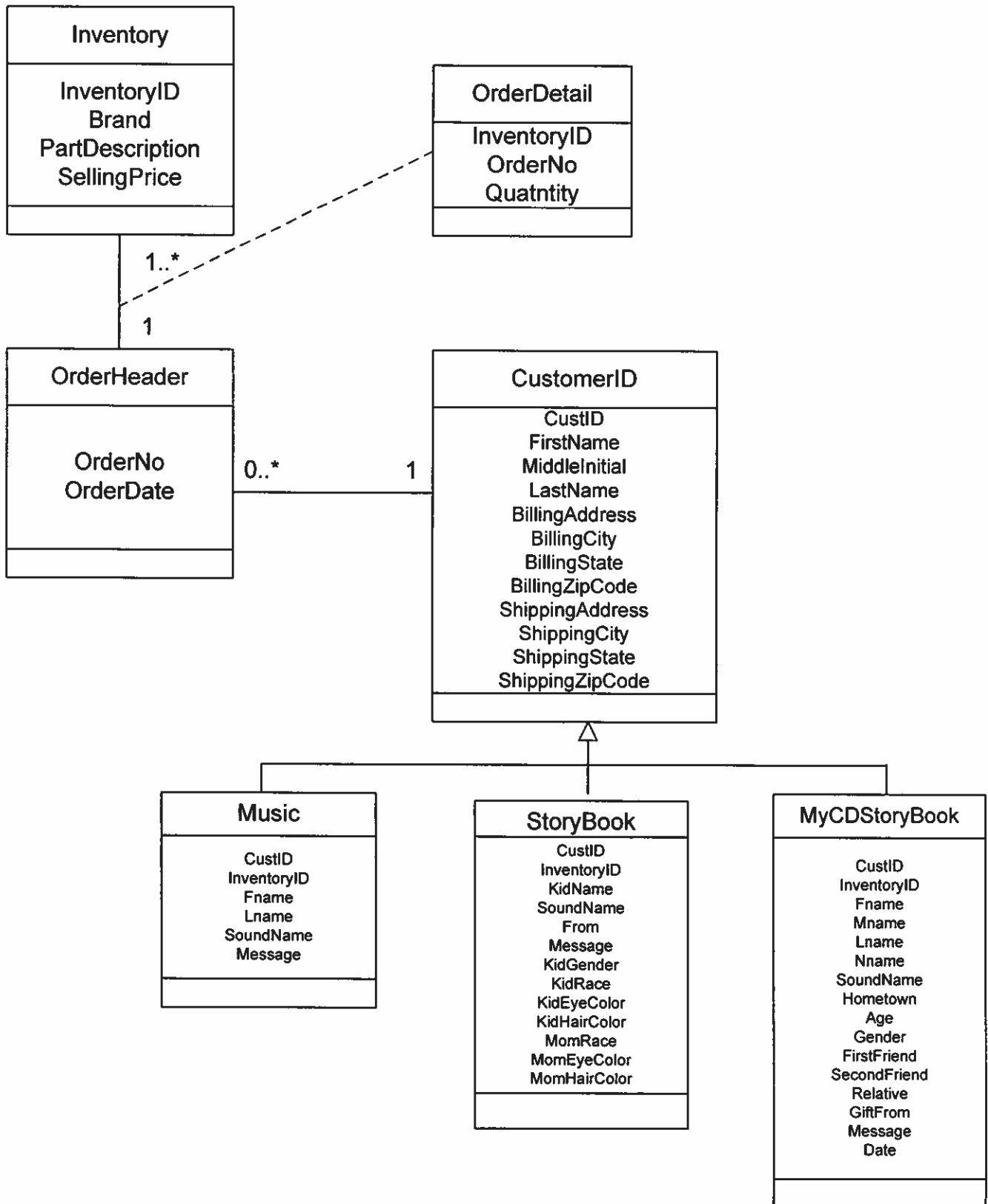
Complete Event Table

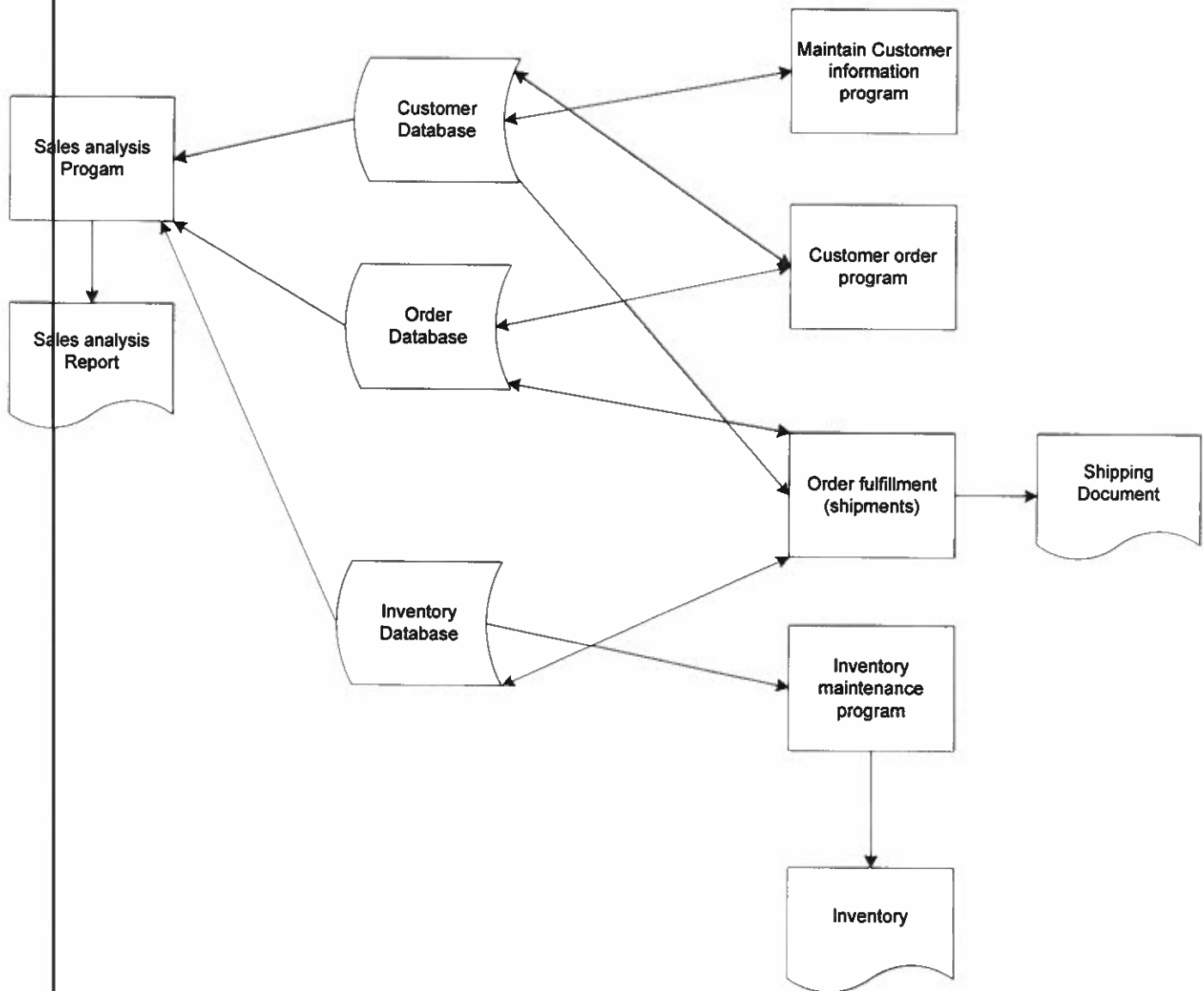
Event	Trigger	Source	Activity/Use Case	Response	Destination
Customer places an order	New Order	Customer	Create new order	Order Confirmation Order Details	Customer Shipping
Customer changes or cancels order	Order change request	Customer	Update order	Change confirmation Order change details	Customer Shipping
Time to produce order summary reports	"End of week, month, quarter, and year"		Produce order summary reports	Order summary reports	Management
Check order status	Order status inquiry	Customer or Management	Look up order status	Transaction summary reports	Accounting
Shipping fulfills order	Order fulfillment notice	Shipping	Record order fulfillment		
Shipping identifies back order	Back-order notice	Shipping	Record back order	Back-order notification	Customer
Customer updates account information	Customer account update notices	Customer	Update customer account		
Marketing wants to send newsletter	Newsletter Details	Marketing	Distribute newsletter	Newsletter	Customer or Prospective Customer

Class Diagram



Class Diagram
Revision Feb 2005





System Flowchart

Kids JukeBox USA

Designed by: Aleisha Vrooman
Class of 2005

Kids JukeBox USA

My topic is to create a working web page for David Scarborough that will incorporate a working database for customer information and order forms.

At the present moment there is a web page designed, but my goals are to edit this working site to be more user friendly, and design a database for Mr. Scarborough to be able to access for his personal business rather than relying on the software through internet companies.

In order for me to complete my project I will need to be able to edit the site, add a search tool for customers, and design and code in the database for his order form and customer information. I feel that I will be able to complete my goals by Spring of 2005.

Statement of Scope and Charter

Problem Description: Kids JukeBox USA knows that in order to remain competitive with personalized CD's for children, they must have a working webpage that allows users to order available merchandise. Specifically, there must be a user friendly website in order for the users to search easily and to order their product with a reliable database. Also, the variety of product and customized systems should be considered for the customer. We need to create a system which lets the user choose from a variety of different CD's with a variety of names to insert into each CD. This system also needs to allow the customer to input their personal information so that shipping and billing is possible. Therefore a working, user friendly, web page needs to be edited, and a working database needs to be created and coded into the web page.

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- Provide accurate inventory data
- Allow customers to browse through the product information and order the product with ease.

Database Details

Here is the list of different tables and what each requires.

Music

1. Child's Name How they want it spelled on the label
2. How the child's name sounds phonetically
3. Special Message/Greeting two lines for example Happy 2nd Birthday Love Mom and Dad

Database Details

StoryBook

1. Child's First Name
2. Child's Last Name
3. How the child's name sounds phonetically
4. Special Message/Greeting two lines (same as above)
5. Child's Gender
6. Child's Race
7. Child's Eye color
8. Child's Hair color
9. Parent's Race
10. Parent's Eyecolor
11. Parent's Hair color

Database Details

MYCD StoryBook

1. First Name
2. Middle Name
3. Last Name
4. Nick Name (to be printed throughout the story if different from above)
5. Name spelled phonetically
6. Hometown
7. Age
8. Gender
9. 1st friend
10. 2nd friend
11. Choose one of the following, Mom, Dad, other
12. Who Gift is from ex Persons the CD is from
13. Dedication ex just for you, Happy Birthday
14. Date gift is given ex April 4, 2004

ERD Revised Descriptions

- The CustomerInfo table relates one to many with the OrderHeader, MyCDStoryBook, StoryBook, and Music tables. It includes a primary key of the CustID number. This table is where the key information about the customer is stored.
- The Music table relates many to one with the CustomerInfo table. It includes a primary key made up of the CustID and InventoryID. This table is where information on the child's name and an optional message from the parent is stored.
- The StoryBook table relates many to one with the CustomerInfo table. It includes a primary key made up of the CustID and InventoryID. This table is where the child's name and information about them is stored. The parent's information is also stored in this table.

ERD Descriptions (cont)

- The MyCDStoryBook table relates many to one with the CustomerInfo table. It contains a primary key made up of CustID and InventoryID. This table contains detailed information on the child in order to create a personalized storybook.
- The OrderHeader table relates many to one with CustomerInfo table and one to one with the OrderDetail table. It contains a primary key of OrderNo. This table stores the order and the date of the order.
- The OrderDetail table relates one to one with the OrderHeader table and one to many with the Inventory table. This table takes the OrderNo and the InventoryID to create a primary key from these two foreign keys. This table stores the quantity of a particular item from a particular order.
- The Inventory table relates many to one with the OrderDetail table, and relates one to many with the Music, StoryBook, and MyCDStoryBook tables. It includes a primary key of InventoryID. This table contains each item and the description of the items listed for sale.

[illegible]

Entity-Relationship Diagram Revised

```

    erDiagram
        Customer ||--o{ Order : "places"
        Order ||--o{ OrderItem : "has"
        OrderItem ||--o{ Inventory : "uses"
        Inventory ||--o{ Supplier : "purchases"

        Customer {
            string LastName PK
            string FirstName
            string Address
            string City
            string State
            string ZipCode
            string PhoneNumber
            string Email
            string Password
            string SecurityCode
            string OrderID FK
        }

        Order {
            string OrderID PK
            string OrderDate
        }

        OrderItem {
            string OrderID PK
            string ItemID PK
            string Quantity
        }

        Inventory {
            string ItemID PK
            string Quantity
            string Location
            string SupplierID FK
        }

        Supplier {
            string SupplierID PK
            string Name
            string Address
            string City
            string State
            string ZipCode
            string PhoneNumber
            string Email
            string Password
            string SecurityCode
            string OrderID FK
        }
  
```

The diagram illustrates the following entities and their attributes:

- Customer** (Primary Key: LastName):
 - Attributes: LastName, FirstName, Address, City, State, ZipCode, PhoneNumber, Email, Password, SecurityCode, OrderID (Foreign Key).
- Order** (Primary Key: OrderID):
 - Attributes: OrderID, OrderDate.
- OrderItem** (Composite Primary Key: OrderID, ItemID):
 - Attributes: OrderID, ItemID, Quantity.
- Inventory** (Primary Key: ItemID):
 - Attributes: ItemID, Quantity, Location, SupplierID (Foreign Key).
- Supplier** (Primary Key: SupplierID):
 - Attributes: SupplierID, Name, Address, City, State, ZipCode, PhoneNumber, Email, Password, SecurityCode, OrderID (Foreign Key).

Relationships are shown as follows:

- Customer to Order:** A one-to-many relationship where one customer can place multiple orders.
- Order to OrderItem:** A one-to-many relationship where one order contains multiple items.
- OrderItem to Inventory:** A one-to-many relationship where one item is used in multiple order items.
- Inventory to Supplier:** A one-to-many relationship where one inventory item is purchased from multiple suppliers.

Original ERD with Primary and Foreign Keys

ERD With Primary Keys and Foreign Keys

Table	Attributes
Customer	CustomerID, Name, Address, City, State, Zip, Birthdate, Gender, Email
Employee	EmployeeID, Name, Address, City, State, Zip, Birthdate, Gender, Email
Product	ProductID, Name, Description, Category, Price
Supplier	SupplierID, Name, Address, City, State, Zip, Birthdate, Gender, Email
Order	OrderID, CustomerID, EmployeeID, ProductID, SupplierID, Quantity, Price, DateOrdered, DateShipped
Inventory	ProductID, SupplierID, Quantity, Price, DateOrdered, DateShipped
Shipment	ShipmentID, OrderID, ProductID, SupplierID, Quantity, Price, DateOrdered, DateShipped
Warehouse	WarehouseID, Name, Address, City, State, Zip, Birthdate, Gender, Email

ERD Revised with Primary and Foreign Keys

ERD With Primary Keys and Foreign Keys

Table	Attributes
Customers	CustomerId, FirstName, LastName, Address, City, State, Zip, Phone, Email, Password, ShippingAddress, ShippingCity, ShippingState, ShippingZip
OrderDetail	OrderDetailId, OrderId, ProductId, Quantity
OrderHeader	OrderId, OrderDate, ShipDate, ShipVia, ShipCost, ShipMethod, ShipAddress, ShipCity, ShipState, ShipZip, ShipCountry, ShipRegion
Product	ProductId, ProductName, ProductNumber, ProductDescription, ProductImage, ProductCategory, ProductSubcategory, ProductManufacturer, ProductSupplier, ProductCost, ProductPrice, ProductStatus, ProductType, ProductUnit, ProductWeight, ProductVolume, ProductLength, ProductWidth, ProductHeight, ProductColor, ProductMaterial, ProductTexture, ProductTaste, ProductSmell, ProductSound, ProductTactile, ProductTaste, ProductSmell, ProductSound, ProductTactile
Shipment	ShipmentId, OrderId, ShipDate, ShipVia, ShipCost, ShipMethod, ShipAddress, ShipCity, ShipState, ShipZip, ShipCountry, ShipRegion

Table: Customers

CustomerId	FirstName	LastName	Address	City	State	Zip
101	Charles	Smith	452 Alpha Ave	Shawnee	KS	66204
102	James	Cameron	120 Beta Drive	Argentine	NC	28000
103	Shirley	Raymond	502 Gamma Ave	Nebraska	NE	68000

CustomerId	FirstName	LastName	Address	City	State	Zip
101	Charles	Smith	452 Alpha Ave	Shawnee	KS	66204
102	James	Cameron	120 Beta Drive	Argentine	NC	28000
103	Shirley	Raymond	502 Gamma Ave	Nebraska	NE	68000

Table: Inventory

InventoryId	CustomerId	Quantity	Price
101	Charles	1	\$19.95
102	James	1	\$19.95
103	Shirley	1	\$19.95

InventoryId	CustomerId	Quantity	Price
101	Charles	1	\$19.95
102	James	1	\$19.95
103	Shirley	1	\$19.95

Table: Music

CustomerId	InventoryId	Quantity	Price
101	101	1	\$19.95
102	102	1	\$19.95
103	103	1	\$19.95

CustomerId	InventoryId	Quantity	Price
101	101	1	\$19.95
102	102	1	\$19.95
103	103	1	\$19.95

Table: MYCD StoryBook

InventoryId	CustomerId	Quantity	Price
101	Charles	1	\$19.95
102	James	1	\$19.95
103	Shirley	1	\$19.95

InventoryId	CustomerId	Quantity	Price
101	Charles	1	\$19.95
102	James	1	\$19.95
103	Shirley	1	\$19.95

InventoryId	CustomerId	Quantity	Price
101	Charles	1	\$19.95
102	James	1	\$19.95
103	Shirley	1	\$19.95

Table: Order Detail

OrderDetailId	CustomerId	Quantity	Price
101	Charles	1	\$19.95
102	James	1	\$19.95
103	Shirley	1	\$19.95

OrderDetailId	CustomerId	Quantity	Price
101	Charles	1	\$19.95
102	James	1	\$19.95
103	Shirley	1	\$19.95

Table: Order Header

[illegible]

Form: Order Detail

Display	Inventory@	Brand	PartDescription	SellingPrice	Quantity
1	101	My Very Own	Music CD	\$19.95	1
2	101	My Very Own	Music CD	\$19.95	1
3	102	Awesome Kids Music	CD	\$19.95	1
4	201	My Pet Dinosaur Storybook	CD	\$24.95	1
5	301	The Fairy	MY CDStorybook	\$24.95	1
6	305	The Fool	MY CDStorybook	\$24.95	1

Form Header

Inventory@	Brand	PartDescription	SellingPrice	Quantity
101	My Very Own	Music CD	\$19.95	1
101	My Very Own	Music CD	\$19.95	1
102	Awesome Kids Music	CD	\$19.95	1
201	My Pet Dinosaur Storybook	CD	\$24.95	1
301	The Fairy	MY CDStorybook	\$24.95	1
305	The Fool	MY CDStorybook	\$24.95	1

Form Footer

Form: Order Header

System Context Diagram

The diagram illustrates the interactions between a Customer, a Customer Support System, and Shipping. The Customer is represented by a rectangle on the left. The Customer Support System is a rectangle on the right. Shipping is a rectangle at the bottom. A thick, double-lined hexagon encloses the four messages between the Customer and the Customer Support System: 'New Order', 'New Return Info', 'Customer Change Order', and 'Back Order Info'. A single arrow labeled 'Shipping' points from the Customer Support System to the Shipping box.

```
graph LR; Customer[Customer] -- "New Order" --> CSS[Customer Support System]; CSS -- "New Return Info" --> Customer; Customer -- "Customer Change Order" --> CSS; CSS -- "Back Order Info" --> Customer; CSS -- "Shipping" --> Shipping[Shipping];
```

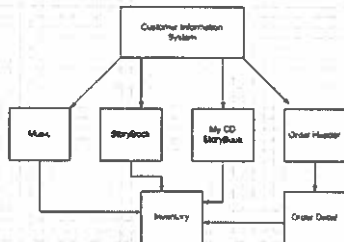
This diagram shows how the customer input information relates back to the support system in order to relay a message about the order back to the customer or directly to shipping to ship the item off.

Structure Chart

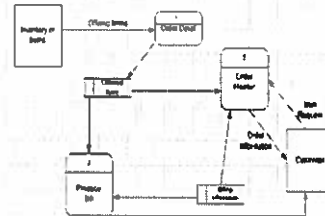
```
graph TD; A[Customer Information System] --> B[Menu]; A --> C[Main Menu]; A --> D[MS SQL Server]; A --> E[Order System]; C --> F[MS SQL Server]; D --> G[MS SQL Server]; E --> H[Order System]; H --> I[Order System];
```

This chart shows a breakdown and structural diagram of the tables in the database. It shows that the Customer Information is the center of the design with each key table listed under it with the tables that are connected to each listed below them. This is my original design of the database, on the next slide you will see the revised edition of this chart.

Structure Chart Revised

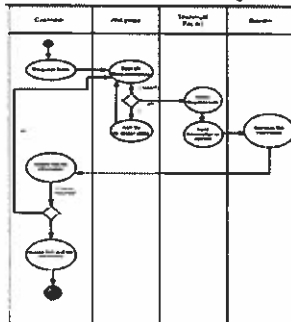


Data Flow Diagram



This diagram shows the flow system of how the customer information relates to the inventory table in order to produce a bill for the customer. The information goes from the Inventory table through the OrderHeader table to the OrderDetail table where the order information is sent to produce a bill and the bill is then sent to the customer.

Activity Diagram



This diagram shows how the system checks the requirements in order to produce a bill. If anything happens with the information that is incorrect, the system goes back to recheck the information. Finally, the system accepts the item and the bill information.

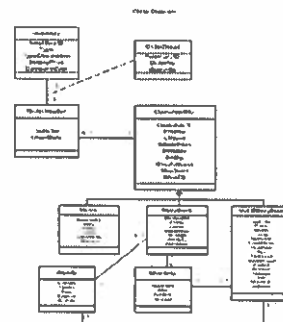
Complete Event Table

Complete Event Table					
Event	Trigger	Source	Activity/Use Case	Response	Destination
Customer places an order	New Order	Customer	Create new order	Order Confirmation Order Details	Customer Shipping
Customer changes or cancels order	Order change request	Customer	Update order	Change confirmation Order change details	Customer Shipping
Time to produce order summary reports	End of week, month, quarter, and year		Produce order summary reports	Order summary reports	Management
Check order status	Order status inquiry	Customer or Management	Look up order status	Transaction summary reports	Accounting
Shipping bill of order	Order fulfillment notice	Shipping	Revised order fulfillment		
Shipping schedule back order	Back-order notice	Shipping	Revised back order	Back-order notification	Customer
Customer updates account information	Customer account update notice	Customer	Update customer account		
Marketing wants to send newsletter	New order details	Marketing	Order data newsletter	Newsletter	Customer or Management Customer

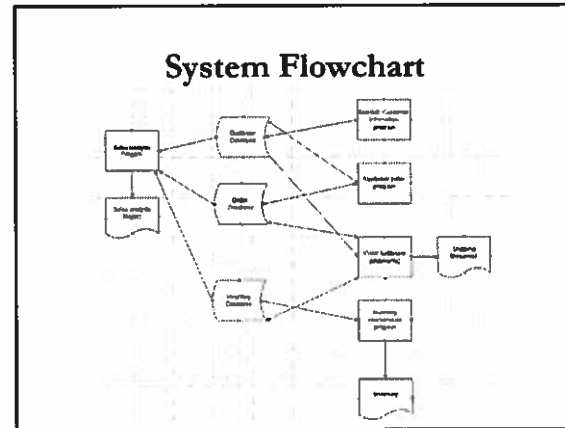
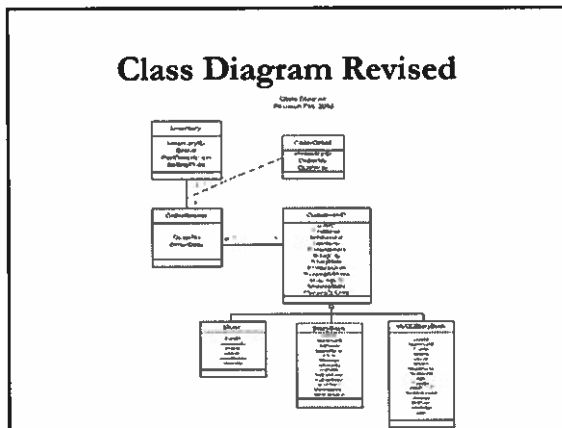
Complete Event Table

This table describes the activities that occur when each event occurs. The source tells where the event started and from there the event causes the activity/use case to occur causing a response from the system sending it to the destination specified.

Class Diagram



This diagram illustrates the flow of the database system along with their relationships towards each other.



System Flowchart

This flowchart describes the direction of the each subsystem in the database. It shows the flow of the data and control among each of the subsystems. It shows the relationship and the flow of data between the customer, the inventory, the order, and the shipping areas of the system.

Inventory Form

Inventory ID	Brand Name	Part Description	Price
101	Star Wars Stormtrooper	Stormtrooper	\$19.95
102	Star Wars Stormtrooper	Stormtrooper	\$19.95
103	Star Wars Stormtrooper	Stormtrooper	\$19.95
104	Star Wars Stormtrooper	Stormtrooper	\$19.95
105	Star Wars Stormtrooper	Stormtrooper	\$19.95
106	Star Wars Stormtrooper	Stormtrooper	\$19.95
107	Star Wars Stormtrooper	Stormtrooper	\$19.95
108	Star Wars Stormtrooper	Stormtrooper	\$19.95
109	Star Wars Stormtrooper	Stormtrooper	\$19.95
110	Star Wars Stormtrooper	Stormtrooper	\$19.95
111	Star Wars Stormtrooper	Stormtrooper	\$19.95
112	Star Wars Stormtrooper	Stormtrooper	\$19.95
113	Star Wars Stormtrooper	Stormtrooper	\$19.95
114	Star Wars Stormtrooper	Stormtrooper	\$19.95
115	Star Wars Stormtrooper	Stormtrooper	\$19.95
116	Star Wars Stormtrooper	Stormtrooper	\$19.95
117	Star Wars Stormtrooper	Stormtrooper	\$19.95
118	Star Wars Stormtrooper	Stormtrooper	\$19.95
119	Star Wars Stormtrooper	Stormtrooper	\$19.95
120	Star Wars Stormtrooper	Stormtrooper	\$19.95

Layout of Music Attributes

Attributes are held on system.

NAME: _____

ARTIST: _____

ALBUM: _____

GENRE: _____

Continue

Layout of StoryBook Attributes

Attributes are held on system.

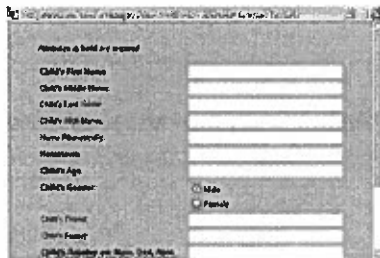
TITLE: _____

AUTHOR: _____

GENRE: _____

Continue

Layout of MyCDStoryBook Attributes



Webpage



Webpage Updates

This semester has been busy with all the webpage edits and updates constantly changing. I have added numerous pages to the webpage, for example:

- Adding a new page for My CD StoryBooks
- Adding new inventory and changing the line-up of all the inventory
- Changing the wording of certain information listed on the main page, charity page, and the faq page

I am always making changes to the site directed by David, the owner of the page. I am currently fixing a few other edit problems and also finishing the database for his webpage.

Webpage Updates

Here is a list of the different requests to update this webpage:

- Compiled one page with a list of songs to listen to & away to order these songs from this page.
- Create a page to list MyCD StoryBook products.
- Update each Music CD with it's complete list of songs that the user can preview.
- Created a new order form button, and links listed on the main page, and a customization button for theStoryBook CD's.
- Make a newsletter form for users to subscribe to a monthly newsletter.
- Create font colors to make it easier to read by the user.
- Created a page to include all the list of songs from each Music CD on one page to make it easier for the user to preview the songs

Finishing Touches

I was able to upload the database to the working webpage. After numerous emails and meetings, David is satisfied with the outcome of his database and webpage. He can now use each product individually, which wasn't possible with his old database that the server provided him.

We have discussed ways to keep his site up to date. I have reviewed with him numerous ways to update things on his webpage himself, and let him know that I will help him in the future if he needs my assistance.

Questions/Comments

At this moment I will take any questions or comments about this presentation and project.